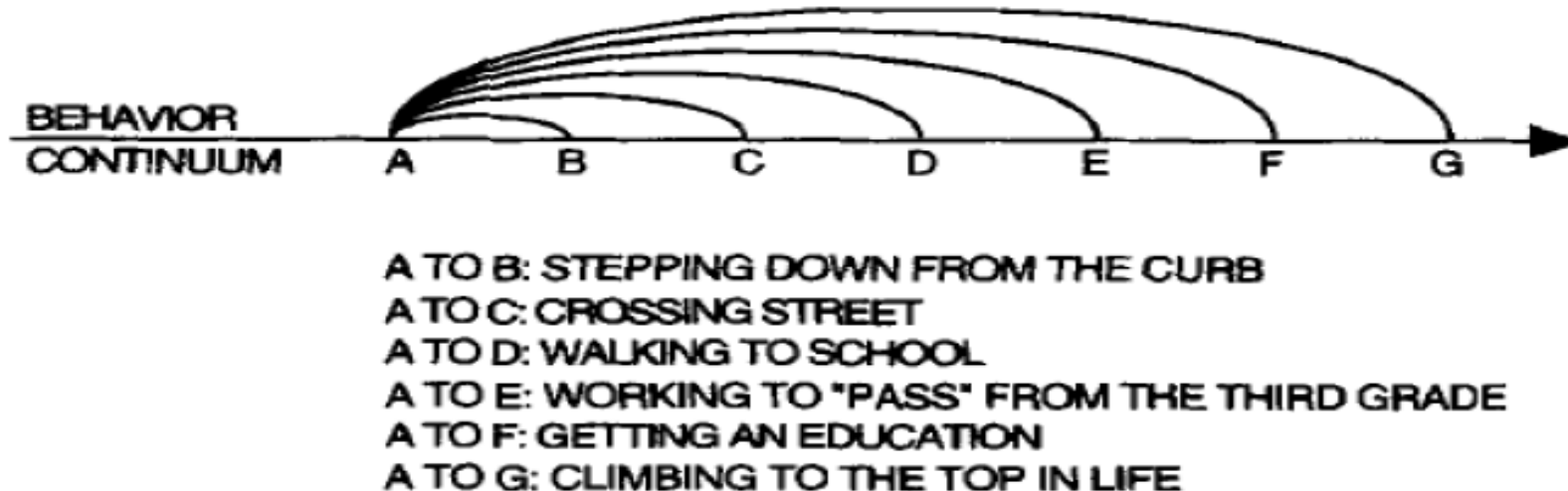


Long horizon planning and the role of language

Jitendra Malik

Hierarchy of human behavior

- The hierarchical structure of human behavior- movement, goals, actions and events e.g. Barker and Wright (1954).



The multiple spatiotemporal scales

- The finest scale can be understood as physical action, but the larger scales are best understood in terms of goals and intentionality

ACTION = MOVEMENT + GOAL

On Mental Models

The image shows the front cover of the book 'The Nature of Explanation' by Kenneth Craik. The cover has a red border. The title 'The Nature of Explanation' is written in a large, white, serif font. Below the title, the author's name 'KENNETH CRAIK' is written in a smaller, white, sans-serif font. The background of the cover is dark blue.

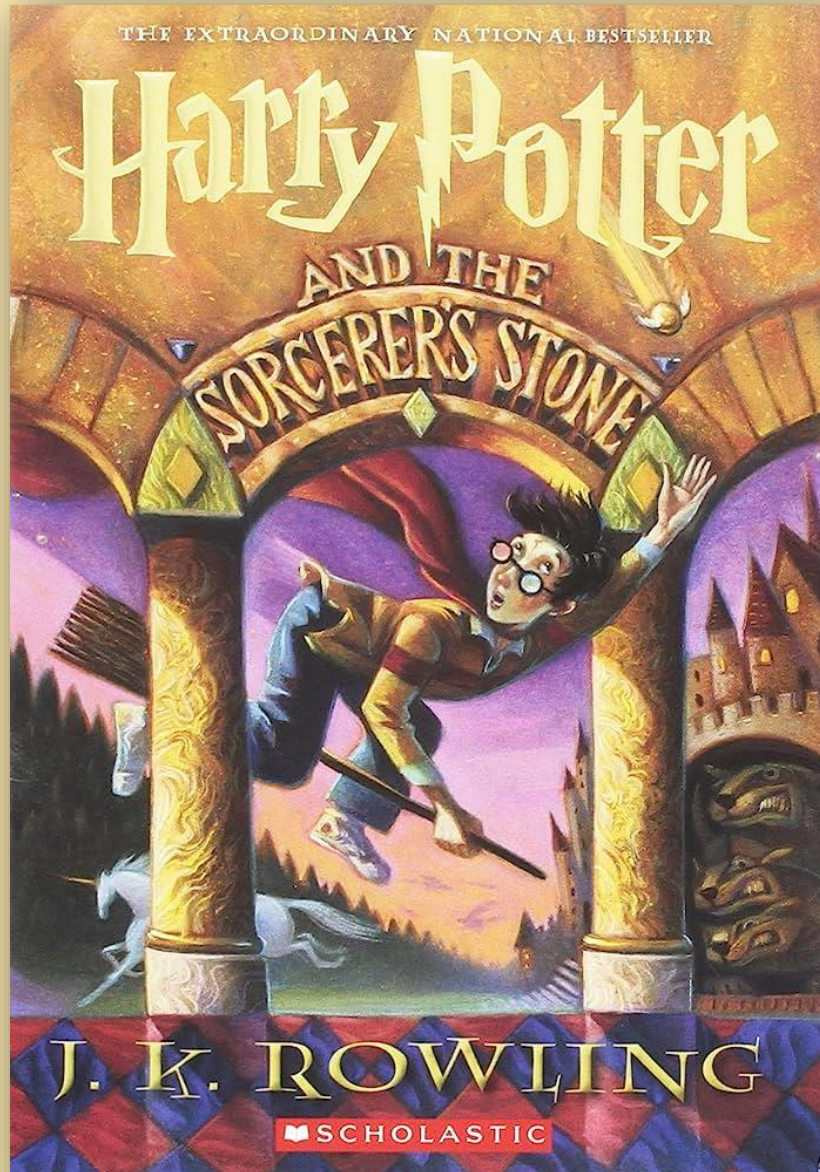
The Nature of Explanation

KENNETH CRAIK

CAMBRIDGE UNIVERSITY PRESS

If the organism carries a 'small-scale model' of external reality and of its own possible actions within its head, it is able to try out various alternatives, conclude which is the best of them, react to future situations before they arise, utilize the knowledge of past events in dealing with the present and the future, and in every way to react in a much fuller, safer, and more competent manner to the emergencies which face it (Craik, 1943, Ch. 5, p.61)

Modern Control theory (Kalman et al) uses a state space formalism to achieve this. Cognitive science notions of schemas/frames are related.



What position does Harry play on the Quidditch team?

- Answer: Seeker

What does the Sorcerer's Stone do?

- Answer: It produces the Elixir of Life which makes the drinker immortal, and can also turn any metal into pure gold.

What is the name of the Mirror which had the Sorcerer's Stone hidden within it?

- Answer: The Mirror of Erised

or we could just watch the movie!



Long horizon planning

- Classic approach going back to SHAKEY (1971). One designs operators with precondition lists, delete lists, and add lists, and then planning is a series of calls to these operators with different arguments.
- There is a sophisticated line of work in the AI planning community building on these ideas. PDDL is used
- TAMP (Lozano-Perez and Kaelbling) is in that vein. The task planning part is in that tradition, the motion planning is specific for robotics problems.
- Language models have implicit knowledge of the operators we might use for executing on tasks such as “Make an omelette”

DOES SPATIAL COGNITION EMERGE IN FRONTIER MODELS?

Santhosh Kumar Ramakrishnan* Erik Wijmans Philipp Krähenbühl Vladlen Koltun

Apple

ABSTRACT

Not yet. We present SPACE, a benchmark that systematically evaluates spatial cognition in frontier models. Our benchmark builds on decades of research in cognitive science. It evaluates large-scale mapping abilities that are brought to bear when an organism traverses physical environments, smaller-scale reasoning about object shapes and layouts, and cognitive infrastructure such as spatial attention and memory. For many tasks, we instantiate parallel presentations via text and images, allowing us to benchmark both large language models and large multimodal models. Results suggest that contemporary frontier models fall short of the spatial intelligence of animals, performing near chance level on a number of classic tests of animal cognition.

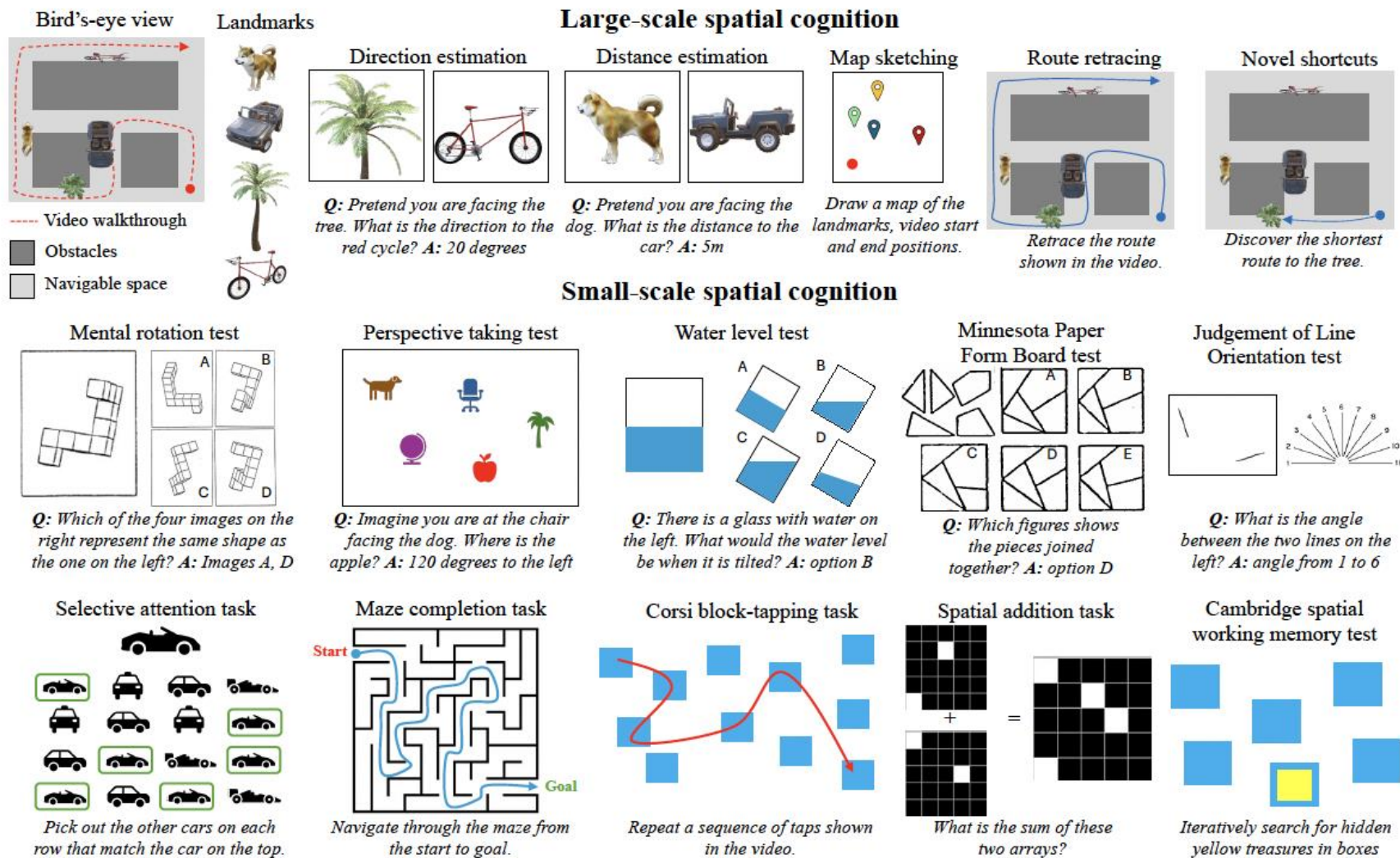


Figure 1: SPACE: Spatial Perception And Cognition Evaluation. We design a suite of spatial cognition tasks based on the cognitive science literature. These are broadly classified into large-scale and small-scale spatial cognition. Large-scale tasks require understanding space at the level of environments, while small-scale tasks require understanding space at the level of objects or object arrangements. We develop multimodal as well as purely textual presentations, which support evaluation of both large language models (LLMs) and vision-language models (VLMs).

Observation space: Ego image						
Method	Direction estimation	Distance estimation	Map sketching	Route retracing	Novel shortcuts	Average
Human	82.8	83.2	96.6	-	-	-
GPT-4o	32.0 $\pm$ 4.1	36.5 $\pm$ 5.0	33.3 $\pm$ 4.1	6.6 $\pm$ 3.6	6.4 $\pm$ 1.0	23.0
GPT-4v	29.7 $\pm$ 0.3	31.9 $\pm$ 2.7	20.0 $\pm$ 11.8	1.6 $\pm$ 1.2	3.9 $\pm$ 0.9	17.4
Chance	25.0	25.0	25.0	0.0	0.0	15.0
Observation space: BEV image						
Method	Direction estimation	Distance estimation	Map sketching	Route retracing	Novel shortcuts	Average
Human	82.9	82.5	100.0	-	-	-
GPT-4o	29.5 $\pm$ 5.5	31.9 $\pm$ 1.0	33.3 $\pm$ 3.3	23.6 $\pm$ 3.1	25.9 $\pm$ 2.0	28.8
GPT-4v	26.3 $\pm$ 3.0	29.3 $\pm$ 4.1	45.0 $\pm$ 5.0	13.7 $\pm$ 5.2	15.3 $\pm$ 3.0	25.9
Chance	25.0	25.0	25.0	0.0	0.0	15.0
Observation space: BEV text						
Method	Direction estimation	Distance estimation	Map sketching	Route retracing	Novel shortcuts	Average
Human	66.7	76.5	66.7	-	-	-
GPT-4o	28.7 $\pm$ 4.1	33.3 $\pm$ 1.7	46.7 $\pm$ 4.1	27.5 $\pm$ 3.2	26.6 $\pm$ 0.1	32.6
Mistral 123B	30.5 $\pm$ 5.1	28.9 $\pm$ 5.7	38.3 $\pm$ 5.5	20.3 $\pm$ 2.8	19.9 $\pm$ 3.0	27.6
GPT-4v	30.7 $\pm$ 4.1	26.5 $\pm$ 2.7	40.8 $\pm$ 6.0	20.6 $\pm$ 5.8	15.4 $\pm$ 2.0	26.8
Llama 3 70B	27.0 $\pm$ 2.2	30.4 $\pm$ 1.9	35.0 $\pm$ 8.3	13.2 $\pm$ 9.2	5.3 $\pm$ 4.1	22.2
Yi 1.5 34B	26.2 $\pm$ 4.7	35.7 $\pm$ 1.4	35.0 $\pm$ 10.7	3.2 $\pm$ 0.2	1.1 $\pm$ 1.6	20.2
Mixtral 8x22B	21.3 $\pm$ 1.9	19.4 $\pm$ 1.4	39.2 $\pm$ 12.6	1.5 $\pm$ 1.4	3.9 $\pm$ 1.7	17.0
Yi 1.5 9B	10.8 $\pm$ 1.0	20.0 $\pm$ 3.7	35.0 $\pm$ 5.0	5.0 $\pm$ 2.2	1.3 $\pm$ 1.5	14.4
Llama 3 8B	22.5 $\pm$ 2.9	24.6 $\pm$ 2.1	23.3 $\pm$ 7.1	0.0 $\pm$ 0.0	1.1 $\pm$ 1.6	14.3
Mixtral 8x7B	15.8 $\pm$ 2.0	16.1 $\pm$ 1.4	30.0 $\pm$ 8.2	1.1 $\pm$ 1.6	1.1 $\pm$ 1.6	12.8
Chance	25.0	25.0	25.0	0.0	0.0	15.0

Table 1: **Large-scale spatial cognition results.** The three tables show results for different observation spaces. Results below 50% of human performance are gray. Methods are sorted based on their overall performance.

INSTRUCTIONS ON HOW TO CLIMB A STAIRCASE

No one will have failed to observe that frequently the floor bends in such a way that one part rises at a right angle to the plane formed by the floor and then the following section arranges itself parallel to the flatness, so as to provide a step to a new perpendicular, a process which is repeated in a spiral or in a broken line to highly variable elevations. Ducking down and placing the left hand on one of the vertical parts and the right hand upon the corresponding horizontal, one is in momentary possession of a step or stair. Each one of these steps, formed as we have seen by two elements, is situated somewhat higher and further than the one prior, a principle which gives the idea of a staircase, while whatever other combination, producing perhaps more beautiful or picturesque shapes, would be incapable of translating one from the ground floor to the first floor.

You tackle a stairway face on, for if you try it backwards or sideways, it ends up being particularly uncomfortable. The natural stance consists of holding oneself upright, arms hanging easily at the sides, head erect but not so

much so that the eyes no longer see the steps immediately above, while one tramps up, breathing lightly and with regularity. To climb a staircase one begins by lifting that part of the body located below and to the right, usually encased in leather or deerskin, and which, with a few exceptions, fits exactly on the stair. Said part set down on the first step (to abbreviate we shall call it "the foot"), one draws up the equivalent part on the left side (also called "foot" but not to be confused with "the foot" cited above), and lifting this other part to the level of "the foot," makes it continue along until it is set in place on the second step, at which point the foot will rest, and "the foot" will rest on the first. (The first steps are always the most difficult, until you acquire the necessary coordination. The coincidence of names between the foot and "the foot" makes the explanation more difficult. Be especially careful not to raise, at the same time, the foot and "the foot.")

Having arrived by this method at the second step, it's easy enough to repeat the movements alternately, until one reaches the top of the staircase. One gets off it easily, with a light tap of the heel to fix it in place, to make sure it will not move until one is ready to come down.

Julio Cortazar

h/t Alyosha Efros

